3_Reading&ExportingCsvFiles

February 11, 2025

[24]:
```python
import pandas as pd
```

[25]:
```python
df = pd.read_csv('stock_data.csv')
```

[26]:
```python
df.head(2)
```

[26]:
```
  Datasource: Google finance Unnamed: 1 Unnamed: 2 Unnamed: 3  Unnamed: 4
0                    tickers        eps    revenue      price      people
1                      GOOGL      27.82         87        845  larry page
```

[29]:
```python
df = pd.read_csv('stock_data.csv', header = 1)
```

[30]:
```python
df.head(2)
```

[30]:
```
  tickers    eps  revenue price      people
0   GOOGL  27.82       87   845  larry page
1     WMT   4.61      484    65        n.a.
```

[31]:
```python
###Custom Column Names
```

[32]:
```python
df = pd.read_csv('stock_data.csv', header=1, names =
 ↪['stock_symbol','eps','revenue','price','people'])
df.head()
```

[32]:
```
  stock_symbol            eps  revenue price         people
0        GOOGL          27.82       87   845     larry page
1          WMT           4.61      484    65           n.a.
2         MSFT             -1       85    64     bill gates
3          RIL  not available       50  1023  mukesh ambani
4         TATA            5.6       -1  n.a.     ratan tata
```

[ ]:
```python
###Limitting the number of rows read
```

[35]:
```python
df = pd.read_csv('stock_data.csv', header = 1, nrows = 2)
df
```

[35]:
```
  tickers    eps  revenue  price      people
0   GOOGL  27.82       87    845  larry page
```

```
1     WMT   4.61      484     65         n.a.
```

[ ]:
```
### Specifying NaNs
```

[37]:
```
df = pd.read_csv('stock_data.csv', header = 1)
df
```

[37]:

	tickers	eps	revenue	price	people
0	GOOGL	27.82	87	845	larry page
1	WMT	4.61	484	65	n.a.
2	MSFT	-1	85	64	bill gates
3	RIL	not available	50	1023	mukesh ambani
4	TATA	5.6	-1	n.a.	ratan tata

[41]:
```
df = pd.read_csv('stock_data.csv', header = 1, na_values = {
             'eps': 'not available',
             'revenue' : [-1],
             'price' : 'n.a.',
             'people' : ['not available','n.a.']})
df
#inside na_values we first input column names and values that we wanna specify␣
 ↪as NaNs
```

[41]:

	tickers	eps	revenue	price	people
0	GOOGL	27.82	87.0	845.0	larry page
1	WMT	4.61	484.0	65.0	NaN
2	MSFT	-1.00	85.0	64.0	bill gates
3	RIL	NaN	50.0	1023.0	mukesh ambani
4	TATA	5.60	NaN	NaN	ratan tata

[42]:
```
df = pd.read_csv('stock_data.csv', header = 1, na_values = ['not available',␣
 ↪-1, 'n.a.'])
df
```

[42]:

	tickers	eps	revenue	price	people
0	GOOGL	27.82	87.0	845.0	larry page
1	WMT	4.61	484.0	65.0	NaN
2	MSFT	NaN	85.0	64.0	bill gates
3	RIL	NaN	50.0	1023.0	mukesh ambani
4	TATA	5.60	NaN	NaN	ratan tata

[44]:
```
df['pe'] = df['price'] / df['eps']
df
```

[44]:

	tickers	eps	revenue	price	people	pe
0	GOOGL	27.82	87.0	845.0	larry page	30.373832
1	WMT	4.61	484.0	65.0	NaN	14.099783
2	MSFT	NaN	85.0	64.0	bill gates	NaN

```
3    RIL      NaN     50.0  1023.0  mukesh ambani         NaN
4    TATA    5.60      NaN     NaN     ratan tata         NaN
```

[48]:
```
df.to_csv('pe.csv', index = False)
```

[ ]:
```

```

[49]:
```
pip install openpyxl
```

```
Collecting openpyxl
  Downloading openpyxl-3.1.5-py2.py3-none-any.whl.metadata (2.5 kB)
Collecting et-xmlfile (from openpyxl)
  Downloading et_xmlfile-2.0.0-py3-none-any.whl.metadata (2.7 kB)
Downloading openpyxl-3.1.5-py2.py3-none-any.whl (250 kB)
Downloading et_xmlfile-2.0.0-py3-none-any.whl (18 kB)
Installing collected packages: et-xmlfile, openpyxl
Successfully installed et-xmlfile-2.0.0 openpyxl-3.1.5
Note: you may need to restart the kernel to use updated packages.
```

[51]:
```
df_movies = pd.read_excel('movies_db.xlsx', 'financials') # financials is the␣
↪name of the sheet we wanna read from the excel file
df_movies.head(4)
```

[51]:

	movie_id	budget	revenue	unit	currency
0	101	1.0	12.5	Billions	INR
1	102	200.0	954.8	Millions	USD
2	103	165.0	644.8	Millions	$$
3	104	180.0	854.0	Millions	Dollars

[53]:
```
###Converting random currency values into USD
def standardize_currency(curr):
    if curr == '$$' or curr == 'Dollars':
        return "USD"
    return curr

df_financials = pd.read_excel('movies_db.xlsx', 'financials', converters = {
    'currency' : standardize_currency
})
df_financials.head()
```

[53]:

	movie_id	budget	revenue	unit	currency
0	101	1.0	12.5	Billions	INR
1	102	200.0	954.8	Millions	USD
2	103	165.0	644.8	Millions	USD
3	104	180.0	854.0	Millions	USD
4	105	250.0	670.0	Millions	USD

[ ]:
```

```